A Neural Shrimp and Its Activity

Stability and Waveform Utility in a Three-Neuron Recurrent Network

Authorship to be finalized by the human researcher

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Private exploratory working draft. EXP-004–007 reproduce and probe a published small recurrent network, including a declared waveform-preservation task. Connectivity is fixed during each trajectory; no training or reasoning benefit is claimed.

Abstract

We reproduce a shrimp-shaped periodic window in the connection-weight plane of a published three-neuron tanh network and measure its consequences for neural activity. A frozen scan evaluates 39,523 grid and transect settings; independent shooting verifies nine- and eighteen-peak orbits. The core waveform repeats every 37.63637538 time units, with dominant transverse multiplier 0.06506. Independent controls show a large difference in output repeatability after matched state perturbations. An initial 4,320-record weight-change experiment suggests a dose-dependent use for atlas guidance, but a stronger follow-up with 4,032 previews and 2,784 target evaluations does not establish a resolved hybrid advantage at new starting weights. At the largest dose, hybrid error is 0.11545 versus 0.26034 for local sensitivity, yet the paired interval includes zero. Tangent and signed-coordinate policies choose identical actions in all 192 conditions; augmented search has lower pooled mean error. Separate continuation retains 196 node records, including 122 unstable records. Numerical failures and unfavorable comparisons remain visible. The shrimp corresponds to actual rhythms and changes in attraction, but periodicity and geometric margin alone do not establish waveform fidelity or a superior control policy. Transfer to trained models remains untested.

1 The measured object

Figure 1 is a map of the neural network itself. Its controls are two connection weights, and its colors describe recurring sequences of the complete three-neuron state. The focused window has a broad nine-peak body and narrow extensions; selected nine- and eighteen-peak trajectories pass independent orbit checks. This is a numerical reproduction of the small-network precedent of da Silva and Rech [2], not a first report of neural shrimps. The term refers to periodic-window organization in a control plane [3]; the corresponding trajectories evolve in state space.

Our question is what this geometry means for the parent network. We therefore place recurrence, sensitivity, neuron potentials, and actual tanh activations side by side. Modern training maps and latent-reasoning basin maps motivate the broader question [4, 5], but the present result concerns the fixed-weight continuous-time system below.

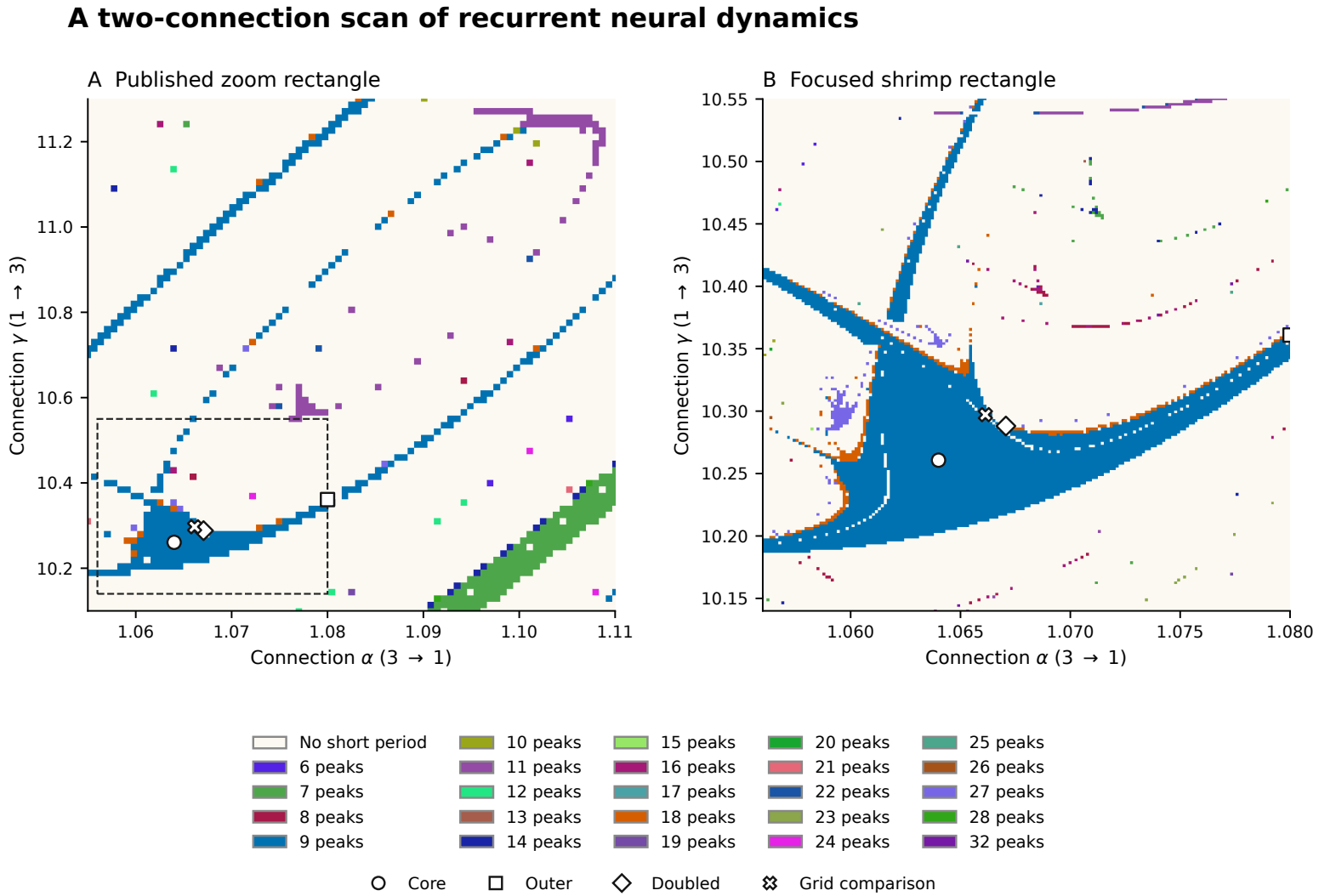


Figure 1: Measured neural parameter map. (A) The published zoom rectangle sampled on an 81×81 grid. (B) The predetermined 181×181 focus resolves the nine-peak shrimp body and extensions, with other periodic candidates nearby. Every cell comes from the neural ODE; there is no drawn shrimp or interpolated reconstruction. Colors count peaks in a recurring *full-state* event sequence, rather than time steps. Pale cells have no detected short period, not a chaos certificate. Markers locate the frozen core, outer-component, doubled, and grid-comparison representatives (the last was originally named exterior). The outer point lies on the focus boundary; its label does not certify a limb tip. The exterior label records coarse-grid selection and fails its intended interpretation under refinement (Table 1).

2 Network, scan, and orbit verification

2.1 The parent network and its readout

The model is the asymmetric three-neuron system in da Silva and Rech [2]:

$$\dot{x} = -x + W\tanh(x), \qquad W = \begin{pmatrix} 1.5 & 2.9 & \alpha \\ -1.5 & 1.18 & 0 \\ \gamma & -22 & 0.47 \end{pmatrix}, \qquad x \in \mathbb{R}^3. \tag{1}$$

The coordinates x_i are neuron potentials, and $u_i = \tanh(x_i)$ are their activation outputs. The weights α and γ respectively couple neurons $3 \to 1$ and $1 \to 3$. They are constant during a run. Thus the experiment changes network connectivity between settings, not a learning rate while training.

2.2 A frozen exploratory reproduction

The overview spans $\alpha \in [1.055, 1.110]$ and $\gamma \in [10.1, 11.3]$ at 81^2 settings. The focus spans $[1.056, 1.080] \times [10.14, 10.55]$ at 181^2 settings. Classical fourth-order Runge–Kutta uses $\Delta t = 0.02$, a settling time of 2,000, and a measurement time of 800. Each increasing-α row begins at $x = (0.1, 0.1, 0.1)$ and then continues from the previous setting's terminal state. Rows are independent. The focused rectangle was selected from the published figure before outcomes existed.

A separate transect fixes $\alpha = 1.064$ and samples 201 values of $\gamma \in [10.14, 10.55]$. Each point independently starts from the same prescribed state, using $\Delta t = 0.01$, settling time 4,000, and measurement time 1,000. It therefore checks whether the main observed structure survives a common initialization without inheriting grid continuation.

A maximum of x_3 defines an event, localized by cubic Hermite interpolation of the numerical endpoints. The retained event includes time and all three state coordinates. The screen searches the last 128 events for four repetitions of a primitive sequence of at most 32 peaks, with absolute tolerance 2×10^{-6} and relative tolerance 2×10^{-5}. A zero label means no short period passed this screen. Finite-time Lyapunov estimates evolve one tangent vector, discard 100 units of alignment, and normalize growth by the remaining measurement time. They are finite-horizon measurements; a stable autonomous cycle has a neutral tangent direction.

2.3 Verification of selected trajectories

The frozen rules select the period-nine candidate nearest $(1.064, 10.26)$, the farthest member of its eight-connected period-nine component, the nearest period-eighteen candidate, and a nearby no-short-period cell with positive finite-time exponent. All four are re-integrated from their captured terminal states at $\Delta t = 0.01$ and 0.005, each with settling time 4,000 and measurement time 1,200. These refinements test the captured regime, not basin independence. Original labels and changed classifications are both retained.

Independent adaptive DOP853 shooting solves $\phi_T(x_0) = x_0$ with an x_3-maximum phase condition, relative state closure at most 10^{-8}, and the expected primitive peak count. Variational integration gives $M(T) = D\phi_T(x_0)$. One multiplier must be neutral and aligned with the flow; both transverse multipliers must have modulus below $1 - 10^{-5}$. Neutral error and flow residual must each be at most 10^{-6}, with alignment at least 0.99. Solver tolerances are 10^{-10} relative and 10^{-12} absolute. These are numerical orbit certificates, not interval-arithmetic proofs.

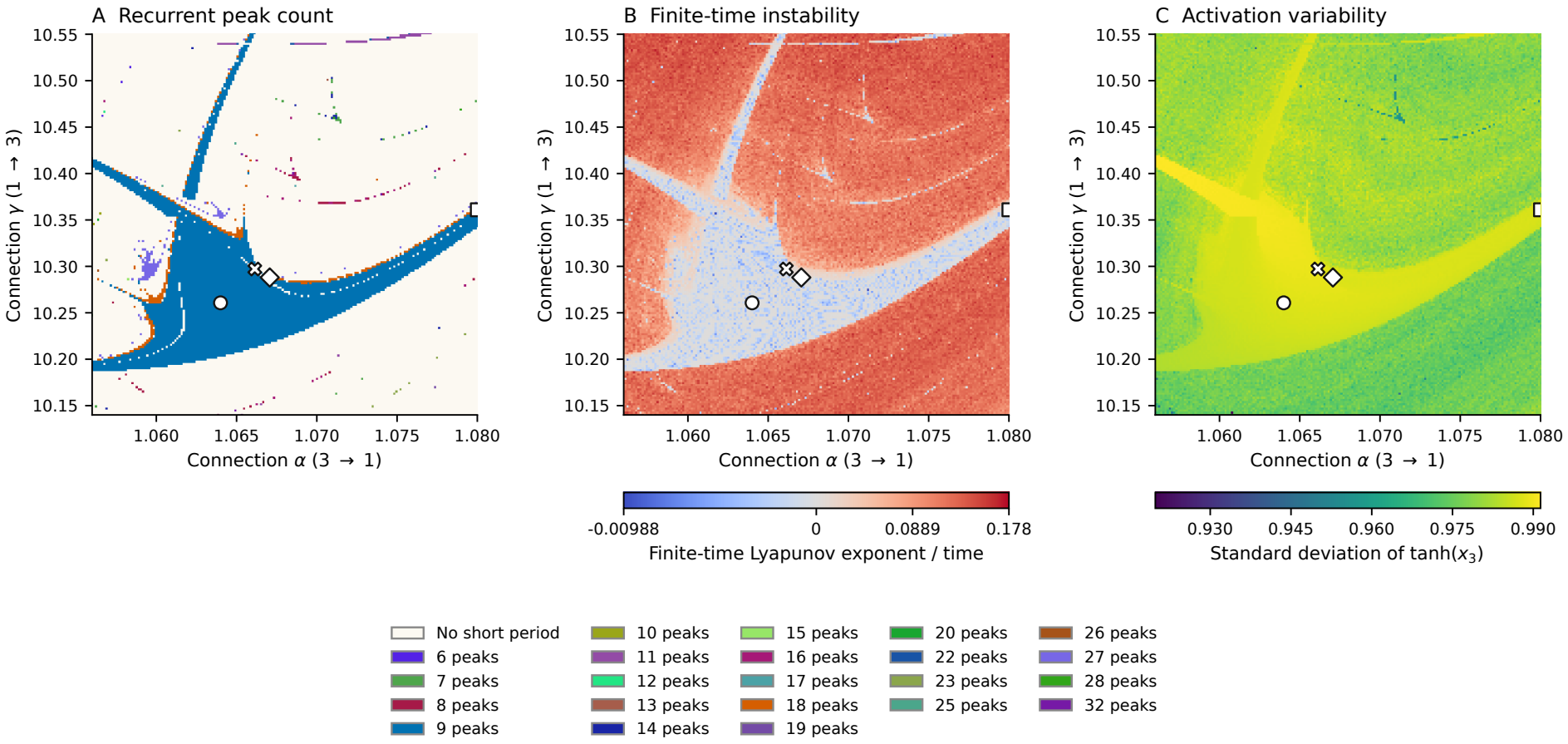


Figure 2: The same geometry in measured network quantities. The focused controls are identical in all three panels: (A) full-state peak recurrence, (B) finite-time Lyapunov estimate per unit time, and (C) standard deviation of neuron 3's actual activation $\tanh(x_3)$. The low-exponent window and activation-statistic structure align visibly with the periodic organization. Panel C uses its labeled narrow range: large color contrast does not imply an equally large absolute activation change. Its statistic summarizes variability and cannot represent the full temporal waveform. A near-zero finite-time exponent in a periodic autonomous flow is consistent with a neutral time direction; positive finite-horizon values alone do not certify asymptotic chaos. All sampled cells and original classifications are retained.

3 How the geometry appears in neural activity

3.1 An independently initialized transect

Figure 3 follows the same connection γ across measured potential maxima, complete-state recurrence, sensitivity, and all three activation variabilities. The recurring branches occupy settings with much smaller finite-time sensitivity than the surrounding irregular sequences. The activation standard deviations change more modestly: the strongest change is in temporal organization, which a single mean or variance cannot fully describe.

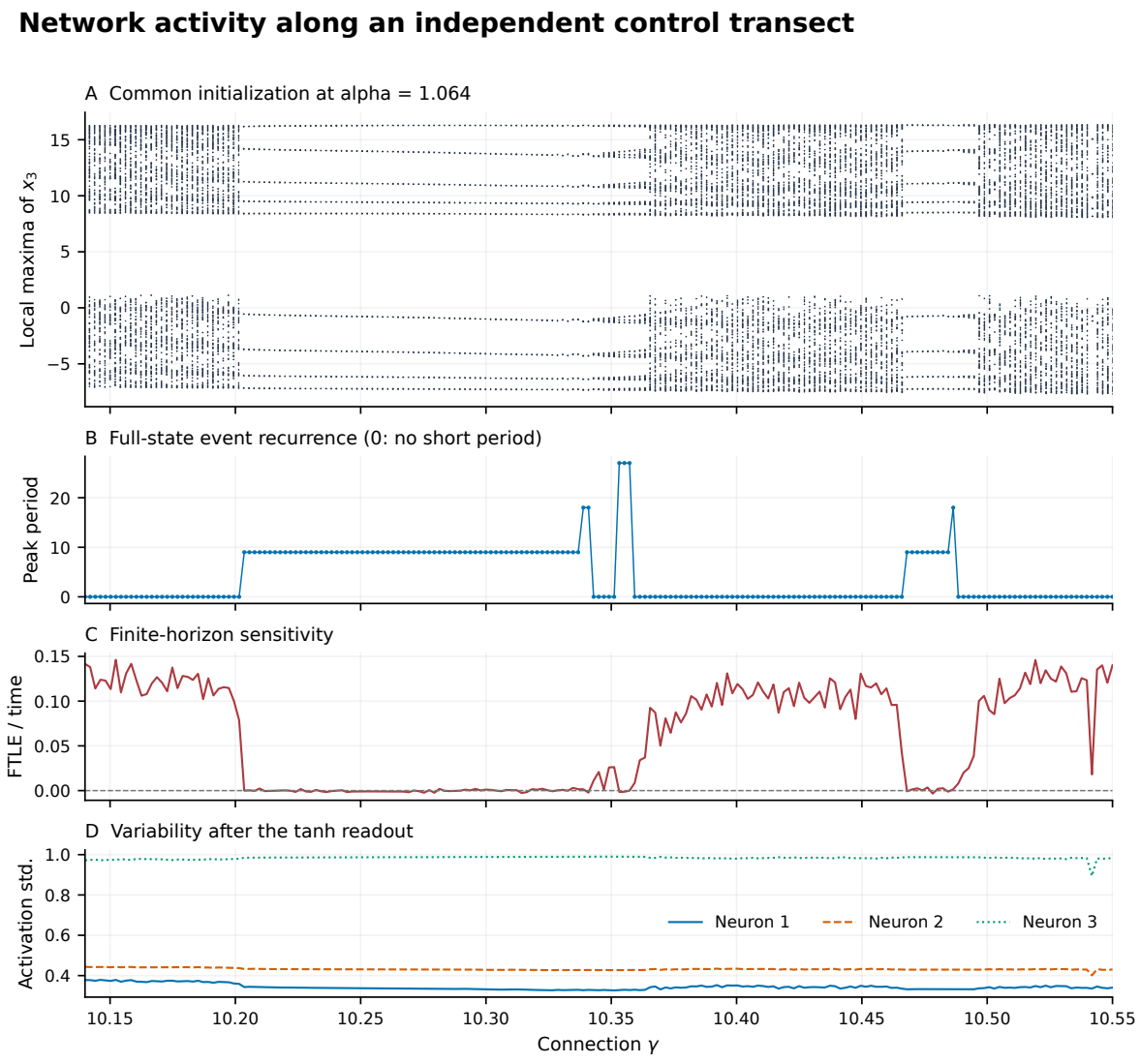


Figure 3: Common-initialization neural transect. At fixed $\alpha = 1.064$, all 201 points independently begin at $(0.1, 0.1, 0.1)$. (A) Every retained local maximum of x_3; (B) candidate peak period from complete-state event recurrence; (C) finite-time Lyapunov estimate; (D) activation standard deviations for the three neurons. A zero period retains the unresolved category. Lines connect measured settings and do not locate exact bifurcation boundaries. This transect uses twice the grid settling time and a smaller step, with no continuation from neighboring controls.

3.2 The full waveform and the effect of saturation

At the verified core, the complete state repeats every 37.63637538 time units, with nine maxima of x_3 in that cycle. The separately integrated behavior probe reports one-period state repetition RMS 8.15×10^{-10} from 2,248 dense-output pairs in its final half. Figure 4 displays every neuron and its output, so the parameter label is connected directly to network motion.

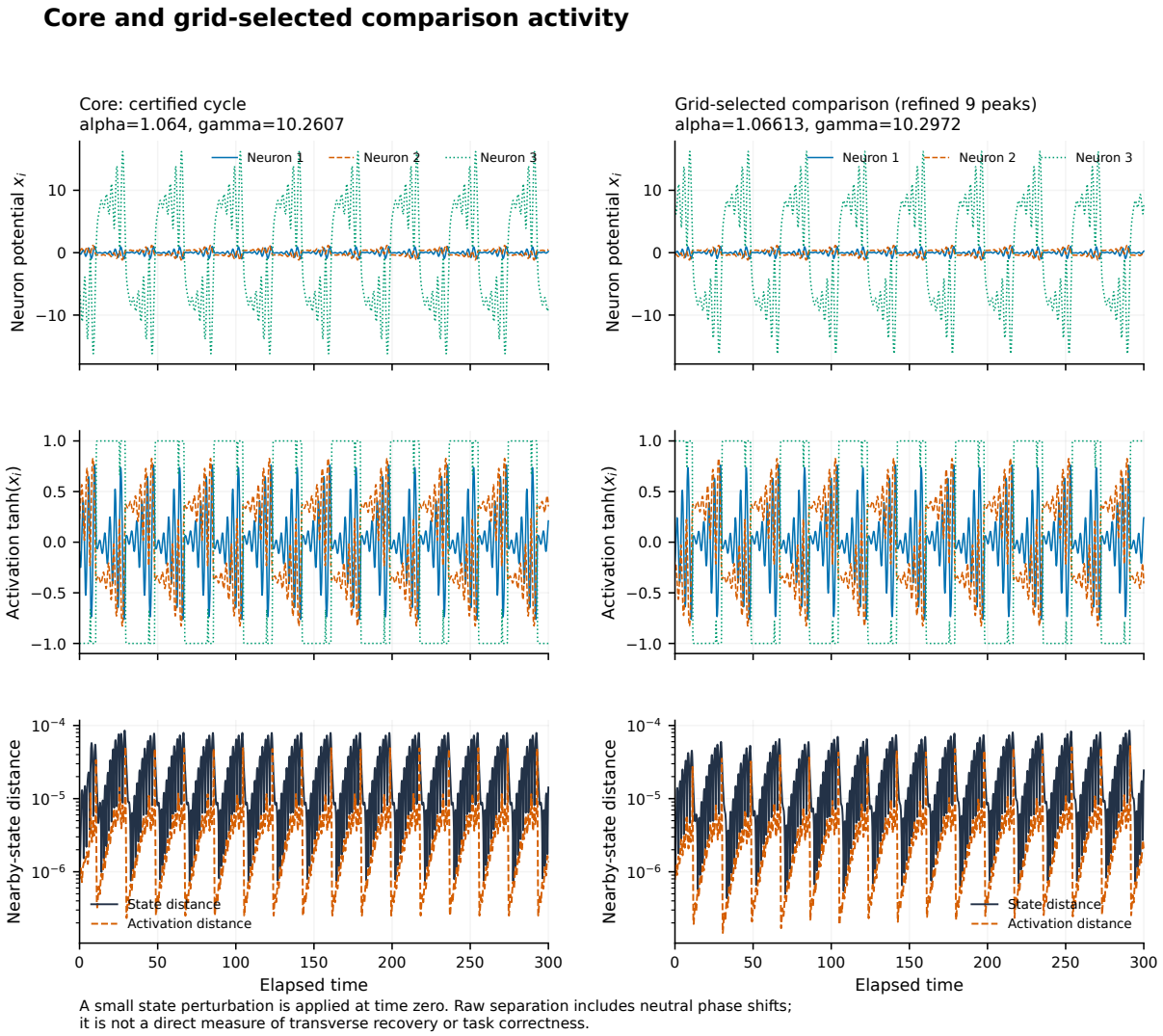


Figure 4: Neuron potentials, activations, and paired perturbations. The core is a numerically verified attracting cycle. The comparison was selected as an exterior by the original grid, but both refinements detect nine peaks; it is *not* a validated chaotic exterior. Columns retain those original roles. Rows show all three potentials, their tanh activations, and same-time state/activation separation after a $10^{-6}(1,1,1)/\sqrt{3}$ perturbation. Every stored time over 300 units is plotted. Persistent or temporarily growing same-time separation can contain a neutral phase shift and transient amplification; it is not a direct measurement of transverse recovery. The failed exterior interpretation is retained rather than replaced by a more favorable example.

The complete activation readout is componentwise injective for finite states: $\tanh(x(t+T)) = \tanh(x(t))$ exactly when $x(t+T) = x(t)$. It therefore preserves the full waveform's primitive period.

Saturation changes amplitude and numerical visibility, not this exact correspondence. At the core, neuron 3's potential spans approximately -16.20 to 16.20 in the recorded behavior window, while its activation approaches ± 1. Over the final 150–300 time units of the 300-unit probe, its population potential and activation standard deviations are respectively 8.664 and 0.9876; these have different units and are reported separately.

4 What survives numerical refinement

4.1 Three independently verified periodic orbits

Table 1 retains the coarse and refined peak counts at every selected coordinate. The core and outer-component point remain nine-peak candidates, and the doubled point remains an eighteen-peak candidate at both smaller steps. Their independently shot periods agree closely with the refined estimates. Full-state closure and the neutral-mode check distinguish these observations from repetition of a scalar loss or a projected trajectory.

Selected role	α	γ	Peaks: .02/.01/.005	Shot T	Closure
Core	1.064000	10.260722	9/9/9	37.636375	7.46×10^{-10}
Outer	1.080000	10.360944	9/9/9	36.974021	1.46×10^{-9}
Doubled	1.067067	10.288056	18/18/18	74.481699	9.69×10^{-11}
Exterior	1.066133	10.297167	0/9/9	—	—

Table 1: Refinement and complete-state closure. Coordinates are rounded for display; exact grid coordinates are retained in the manifest and selected evidence. Closure is $\|\phi_T(x_0) - x_0\| / \max(1, \|x_0\|)$. Zero is no detected short period. The exterior failed the frozen requirement that the original and both refined peak counts agree, so no shooting certificate was issued for it. It remains part of the result.

Orbit	Neutral flow multiplier	Transverse multiplier 1	Transverse multiplier 2
Core	1.000000015	0.06506010	6.15×10^{-11}
Outer	1.000000001	-0.46513914	-7.81×10^{-12}
Doubled	1.000000080	0.65612133	-3.45×10^{-19}

Table 2: All three numerical Floquet multipliers. The neutral direction is retained. Attraction refers to the two transverse directions. Values near zero are numerical approximations with ill-conditioned tiny magnitudes; their final digits and signs should not be interpreted as precisely resolved. Contraction factors are per complete cycle, whose duration differs across rows, not directly comparable rates per unit time.

The core's dominant transverse multiplier is 0.06506, while the outer point's is -0.46514. These describe different local perturbation responses even though both have nine peaks. They do not estimate basin volume, noise tolerance, or recovery from an arbitrary disturbance. The selected outer point reaches the right edge of the focus rectangle, so it cannot establish where that extension terminates. A nearby eighteen-peak orbit is consistent with doubled periodic organization, but this experiment does not continue an exact bifurcation branch connecting the two.

4.2 The exterior control exposes a finite-horizon error

The coarse-grid exterior has no detected short period and a finite-time exponent of 0.00558. Both refinements instead detect a nine-peak sequence. Its captured behavior therefore cannot support

an inside-versus-chaos claim. The change could reflect longer settling, numerical refinement, or their combination; the design changes both and does not isolate the cause. This failure is especially relevant near window boundaries, where a finite map can mistake a long transient for persistent irregular dynamics.

The perturbation traces also require the flow interpretation. At the core, a 10^{-6} perturbation reaches maximum same-time state separation 8.53×10^{-5} over 300 units, despite the attracting transverse multipliers. A neutral phase displacement and transient amplification can coexist with orbital attraction. The saved nearest-orbit distances include finite orbit-sampling error and are not exact phase projections. An inference of useful disturbance rejection requires a dedicated phase-aware measurement.

5 Independent controls show a large activity contrast

EXP-005 is a prospective follow-up chosen after the EXP-004 failure and frozen before its own outcomes. It fixes the earlier core, the two endpoints of the independent transect, and the failed near-edge comparison. Each point independently restarts at $(0.1, 0.1, 0.1)$ at both $\Delta t = 0.01$ and 0.005, settles for 10,000 time units, and is measured for 2,000. The finite-time exponent uses the final 1,900 units after tangent alignment. The two distant controls meet the frozen irregularity rule at both steps: no short recurrence and exponent above 0.005. This supports persistent irregularity over the tested horizons, not an asymptotic-chaos proof.

The core again has a verified nine-peak orbit: $T = 37.63637538$, relative closure 8.55×10^{-11}, and transverse spectral radius 0.06506. The failed near-edge point also receives an independent nine-peak certificate, with $T = 37.33149959$, closure 6.46×10^{-10}, and transverse spectral radius 0.85603. EXP-005 requires agreement between its two fresh runs; it does not require agreement with the original coarse grid. Its new certificate therefore does not erase the EXP-004 failure.

From each finest-step terminal state, the same 10^{-6} state perturbation is applied in direction $(1, 1, 1)/\sqrt{3}$. Figure 5 and Table 3 show the resulting 300-unit probes. The core's final activation separation is 1.53×10^{-6}, versus 1.41 and 2.13 at the lower and upper controls; their maximum state separations reach 29.67 and 31.87. These are actual changes in the network's outputs, with a contrast of about six orders of magnitude at the final checkpoint. They measure same-time repeatability for one declared direction and phase per setting, not population robustness, exact phase-aligned recovery, or useful task performance.

Control	γ	Peaks: .01/.005	FTLE: .01/.005	$\|\Delta x(300)\|_2$	$\|\Delta u(300)\|_2$
Core	10.260722	9/9	$-0.000410/-0.000411$	1.66×10^{-5}	1.53×10^{-6}
Lower	10.140000	0/0	0.138663/0.132212	9.850	1.408
Upper	10.550000	0/0	0.122586/0.135211	16.137	2.129
Failed near-edge	10.297167	9/9	0.001239/0.001239	3.24×10^{-5}	3.42×10^{-6}

Table 3: All four EXP-005 controls. $\alpha = 1.064$ except the retained near-edge coordinate, $\alpha = 1.0661333333$. FTLE is per time unit over 1,900 units; separation uses a distinct 300-unit paired probe. Zero means no detected short period.

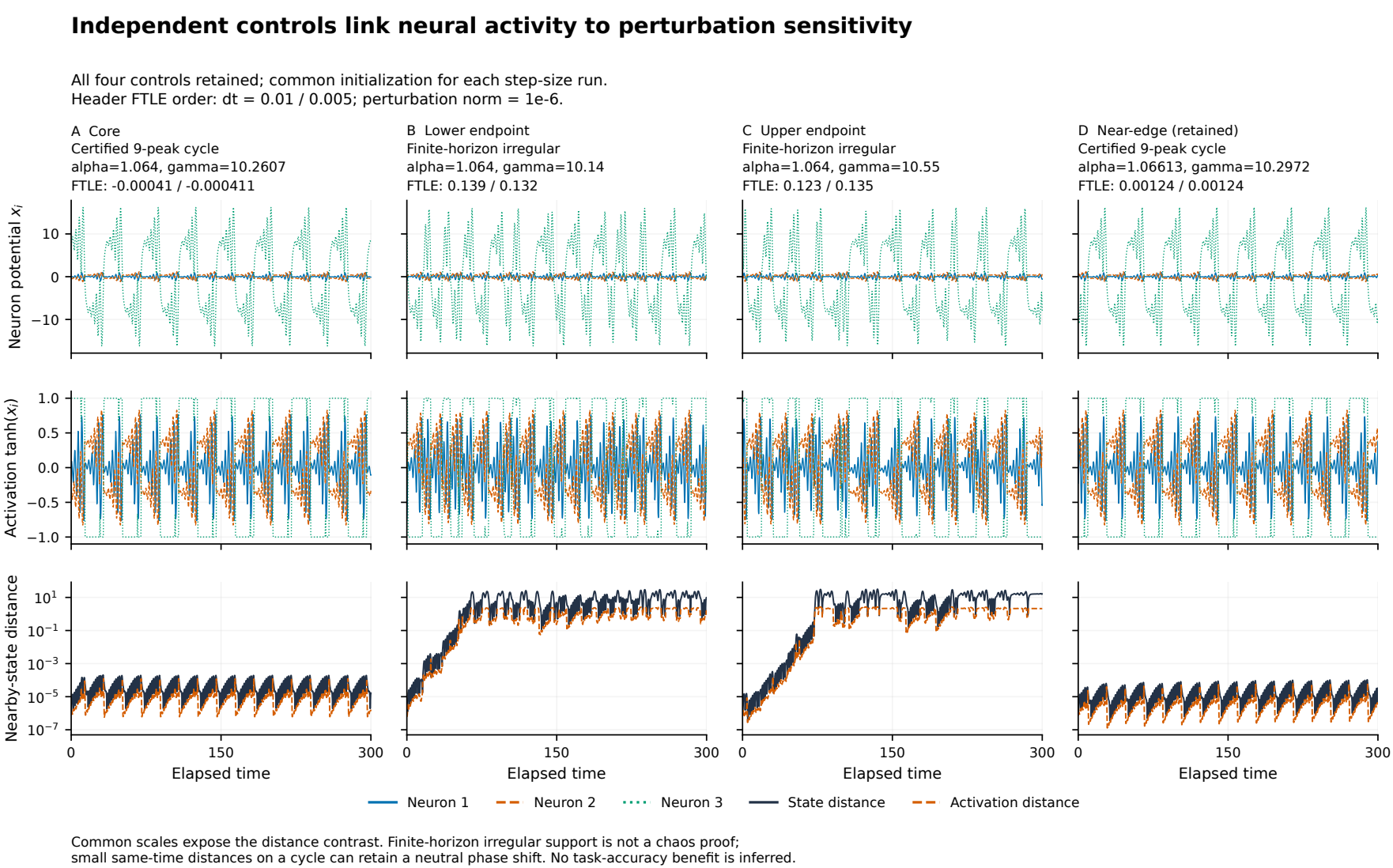


Figure 5: Independent long-settling controls: the parent network behaves differently. EXP-005 retains all four settings: core, lower and upper transect endpoints, and the failed near-edge comparison. Rows show the three neuron potentials, their actual tanh activations, and paired state/activation separations over 300 units. The two distant controls remain irregular by the prespecified finite-horizon rule at both integration steps. Both other settings have independent attracting-cycle certificates. The near-edge column records a new experiment at the failed coordinate; Figure 4 and Table 1 preserve its earlier outcome. All columns use the same perturbation norm and direction; initial phase is the independently obtained terminal state.

6 Testing whether geometry preserves the neural signal

6.1 From peak retention to waveform utility

EXP-004–005 document numerically verified periodic neural activity and motivate a more demanding question: does the measured parameter geometry help preserve a specified output? Retaining nine peaks partly repeats the criterion used to construct the atlas. EXP-006 instead treats the complete activation waveform $u(t) = \tanh x(t)$ as the signal to preserve after a connection-weight change. This is an explicitly defined task in the fixed-weight three-neuron network, not evidence of transfer to trained models, reasoning, or generalization.

6.2 A prospectively fixed waveform-preservation experiment

The exploratory design, chosen after EXP-004–005, fixes twelve anchor attempts from the original nine-peak component, four from each third of its ranked native-coordinate pixel margins. Crop-limited candidates are excluded prospectively; selection uses the fixed seed 20260911 and deterministic maximin coverage. Each anchor must pass fresh common-initialization integrations at steps 0.01 and 0.005, each with 10,000 settling and 2,000 measurement time units, and an independent attracting nine-peak shooting certificate. Failed anchors remain in the census without replacement.

Eligible anchors supply four quarter-cycle initial states shared across arms. The perturbation is applied immediately and held constant. Its radius is one of 0.00025, 0.001, 0.004, or 0.016. Because only $W_{13} = \alpha$ and $W_{31} = \gamma$ change,

$$\|\Delta W\|_F = \sqrt{(\Delta\alpha)^2 + (\Delta\gamma)^2} = r. \tag{2}$$

Thus competing directions have exactly matched native weight norms. The primary atlas policy maximizes the destination's observed component margin over 512 frozen angular candidates, using no new target outcomes. Pixel and crop limitations remain explicit; these margins are not exact bifurcation-boundary distances.

The primary comparator uses local parameter sensitivity, without the atlas:

$$\dot{S} = J(x)S + B(x), \qquad S(0) = 0, \qquad B_{11} = \tanh x_3, \quad B_{32} = \tanh x_1, \tag{3}$$

with other entries zero. Activation sensitivity is $A = \mathrm{diag}(1 - \tanh^2 x)S$. After projecting its sampled columns off the single global activation time-shift vector, the minimum eigenvector of their 2×2 Gram matrix selects a native-unit direction. Errors from its two signs are averaged; the better sign is never selected from outcomes. Eleven direction arms also retain signed normals and tangents, signed random directions, and deranged-anchor normals, alongside a zero-change sham.

The primary error numerically approximates

$$E = \min_{\tau \in [0,T)} \left[\frac{1}{3N} \sum_{k=0}^{N-1} \|u_{\Delta W}(t_k) - u_0(t_k + t_{\mathrm{phase}} + \tau)\|_2^2 \right]^{1/2}, \tag{4}$$

over 300 physical time units sampled every 0.1, including endpoints. The certified reference retains its original period T. One constant phase offset serves the entire record, found by 128 grid candidates and bounded refinement; frequency fitting, time warping, and per-cycle alignment are forbidden. The primary step is 0.005; 0.01 is a separate sensitivity check. Independently initialized long assays measure peak retention, period, recurrence, and finite-time expansion as secondary mechanisms.

The complete census is $12 \times 4 \times (4 \times 11 + 1) \times 2 = 4320$ trial-step records, including ineligible-anchor records. Contrasts subtract atlas error from symmetric-local, symmetric-random, and shuffled-inward error. Equal averaging proceeds over phases, radii, and eligible anchors. The fixed 10,000

anchor-bootstrap draws, seed 20260911, give descriptive percentile 95% intervals, not population inference. Missing required eligible conditions make affected contrasts undefined. Coverage always reports all twelve attempts.

Separately, four pseudo-arclength branches continue two previously certified orbits in both initial γ directions at fixed α, retaining any unstable segments and failed corrections. Transverse Poincaré spectra supply numerical crossing candidates, not exact boundary proofs; continuation never selects utility anchors or actions. Atlas construction and local-sensitivity costs are reported separately, precluding a free-map or net-speedup interpretation.

6.3 The selected network settings all qualify

All twelve attempted anchors pass both new nine-peak screens and independent attracting-orbit certification. Thus none is replaced or omitted. Figure 6 places every anchor on the original measured shrimp. The old map is selection evidence; the qualification checks and interventions are new measurements. Among the 48 frozen anchor–radius atlas choices, twelve selected pixel-margin scores are crop-censored and none uses the all-zero fallback. The experiment evaluates this finite-atlas policy, not an ideal policy given the unknown exact boundary.

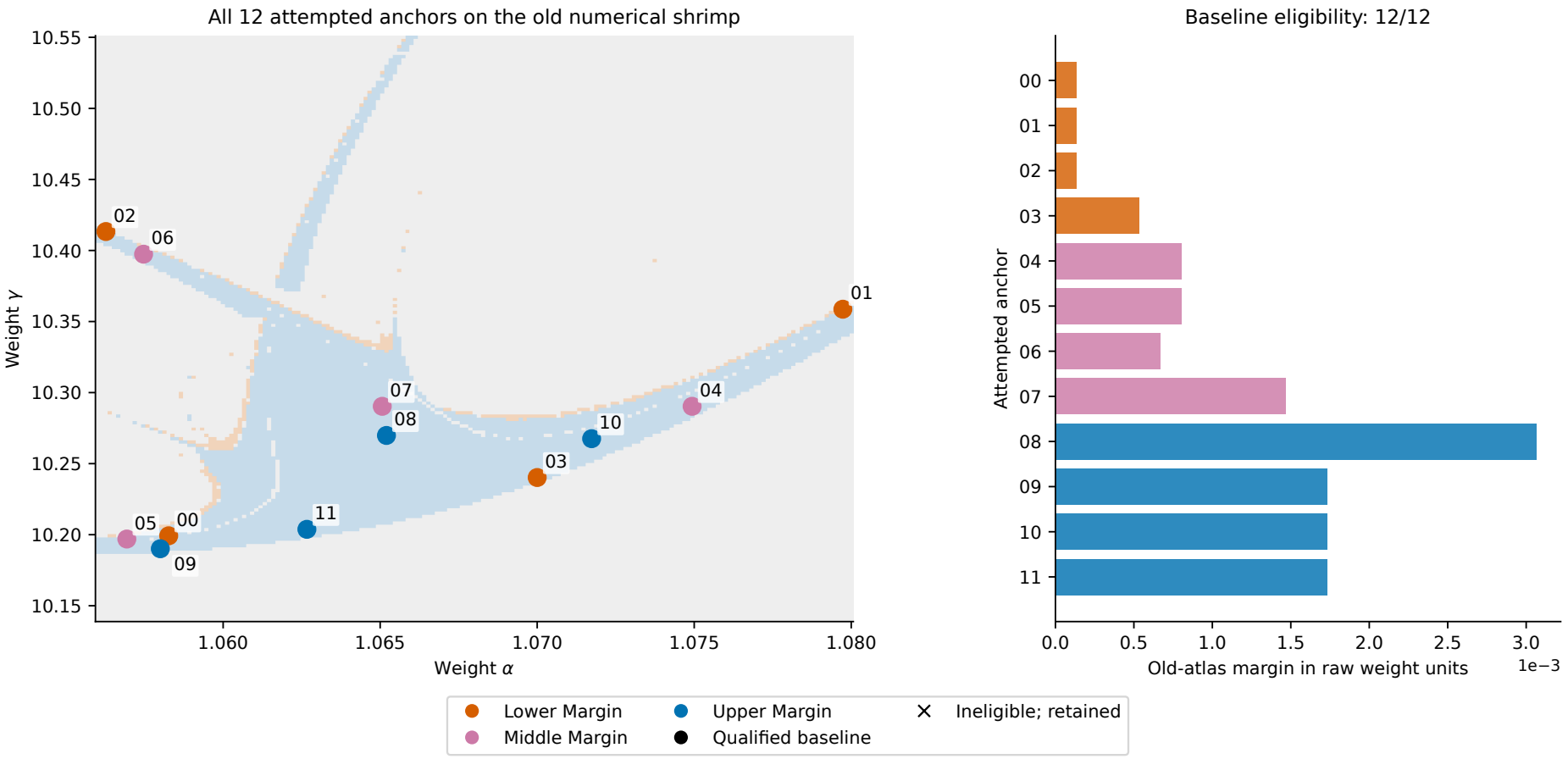


Figure 6: Every attempted utility anchor on the actual neural shrimp. The background is the unchanged EXP-004 numerical atlas. Colors of anchor markers and bars indicate thirds of ranked raw-coordinate pixel margins, not different neurons. All twelve anchors qualify in new independently initialized integrations at both smaller steps and in shooting checks. Margins use native connection-weight units and exclude crop-limited anchor candidates; selected destination scores can still be crop-censored. The complete fixed selection and all 512 angular scores per anchor–radius pair are retained.

6.4 The geometry marks actual changes of orbital stability

Continuation retains 196 node records across four branches, including repeated seeds and overlapping continuations, with 122 unstable node records. These are not 196 distinct attractors. All branches reach their prospectively fixed limit of 48 successors; five unsuccessful correctors among 197 attempts remain in the evidence. Twenty-seven of 28 independent shooting checkpoints pass their agreement checks (Table 4). A reconstruction of the Poincaré spectrum using a different orthonormal section basis agrees with the recorded multipliers to maximum absolute discrepancy 7.11×10^{-14}, without reintegrating the dynamics.

Seed	Initial γ direction	Nodes	Unstable	Failed correctors	Checks passed
Core	$-$	49	31	2/50	7/7
Core	$+$	49	31	0/48	7/7
Near-edge	$-$	49	19	1/49	7/7
Near-edge	$+$	49	41	2/50	6/7

Table 4: Complete continuation census. The direction is the initial orientation; the parameter can reverse as arclength increases. The core and near-edge seeds are the previously certified EXP-005 coordinates. Counts include both copies of each seed and any overlapping orbit segments. The failed independent checkpoint is retained, as are failed corrector attempts.

Along the core's initially increasing-γ branch, a transverse multiplier changes from 0.89961 to 1.26831 between adjacent nodes at $\gamma = 10.32555292$ and 10.32963648. The nine-peak orbit continues beyond this loss of attraction. The initially decreasing branch reverses parameter direction near its sampled minimum $\gamma = 10.20236515$, with a neighboring $+1$ crossing. The near-edge initially increasing branch reverses twice, crosses $+1$ in both directions, and later has a transverse multiplier change from -0.343973 to -1.649692 between adjacent nodes at $\gamma = 10.29183569$ and 10.29404140. This last pair supplies a numerical period-doubling candidate; the newborn branch is not continued here.

These are brackets along continuation arclength. Near a turning point, adjacent nodes' γ values need not bound the crossing's true γ coordinate. Figure 7 therefore reports the continued curves and sampled crossing candidates, not exact bifurcation locations or a complete two-dimensional boundary. A cycle can survive as an unstable orbit while forward trajectories stop approaching it. Orbital existence, orbital attraction, and which basin the initialized network reaches are distinct physical questions.

The independent checkpoint at near-edge positive-branch node 40, $\gamma = 10.38315183$, reports closure 1.05906×10^{-8}, exceeding the frozen 10^{-8} requirement. The continuation calculation reports 1.38×10^{-12} there and both methods agree on state, period, and an unstable transverse radius about 62.635. The failed check is not promoted to a certificate. Compact continuation nodes retain monodromy and gate diagnostics but omit integrated endpoints; the separate audit recomputes spectra, phase and neutral-flow residuals, while closure and primitive-event checks rely on the recorded gates and shooting checkpoints.

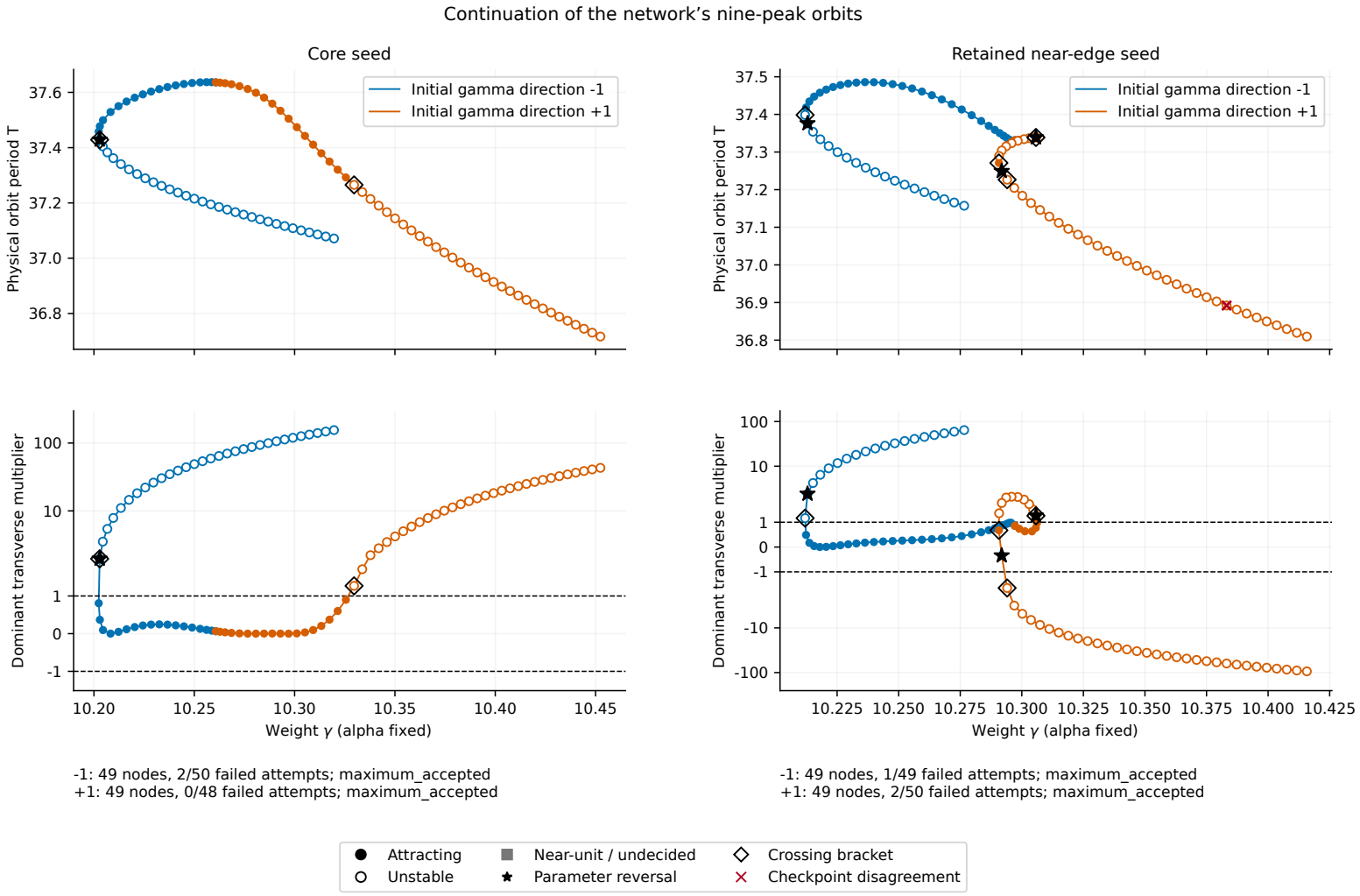


Figure 7: Neural periodic orbits continue through changes in stability. Each column holds one fixed-α seed and both initial continuation orientations. Top panels show physical period versus the actual weight γ; lower panels show the dominant transverse multiplier, with both transverse multipliers retained in raw evidence. Stable and unstable segments, reversals, sampled crossing candidates, and the failed independent checkpoint remain visible. All four branches stop at their node budget. A corrector failure is not interpreted as orbit destruction, and these local sections do not certify the entire shrimp boundary.

6.5 Geometry helps larger changes, while local sensitivity wins nearby

All 4,320 planned trial-step records complete without a failed condition. At the primary step 0.005, the result depends strongly on intervention size (Table 5; Figure 8). At $r = 0.00025$ and 0.001, the symmetric local method has lower mean error at every anchor. At $r = 0.016$, atlas error is 0.301672, versus 0.466653 for the symmetric local method: a 35.35% reduction in mean RMSE. The paired benefit is 0.164981, with descriptive anchor-bootstrap interval $[0.013853, 0.326023]$, and eight of twelve anchor effects favor the atlas. This does not justify claiming an advantage at every setting or dose.

Weight radius	Atlas RMSE	Local RMSE	Local minus atlas	Descriptive 95% interval
0.00025	0.010773	0.000422	−0.010351	[−0.014902, −0.006086]
0.00100	0.036397	0.001725	−0.034672	[−0.056315, −0.015611]
0.00400	0.072170	0.055165	−0.017005	[−0.093829, 0.069591]
0.01600	0.301672	0.466653	0.164981	[0.013853, 0.326023]
Equal-dose mean	0.105253	0.130991	0.025738	[−0.027862, 0.083081]

Table 5: All declared doses, with the local sensitivity comparator. RMSE is the immediate 300-unit, globally phase-aligned three-activation error. Local error averages the two signs of the minimum-sensitivity direction. Each dose averages four phases at each of twelve anchors; the final row weights doses equally. Positive differences favor the atlas. Intervals resample the twelve fixed anchors and are descriptive, not population confidence statements.

The across-dose comparison is unresolved: atlas and local mean errors are 0.105253 and 0.130991, but the interval for their paired difference includes zero. The atlas has lower pooled error than symmetric random directions (0.303602) and shuffled inward normals (0.350651), with positive descriptive benefit intervals $[0.148079, 0.245104]$ and $[0.184833, 0.304101]$. A favorable comparison with random directions therefore does not replace the stronger local comparison. The finite geometry contains useful information for some larger jumps, while a derivative calculated on the original orbit is substantially more accurate for small changes.

The atlas policy is also not the best observed direction arm. The fixed tangent-plus diagnostic has lower mean error than atlas margin at every displayed dose, including 0.260795 versus 0.301672 at the largest dose. This does not promote a diagnostic chosen after inspection to the primary policy. It identifies a concrete competing geometric rule for a new held-out experiment, alongside simple coordinate directions and a stronger local predictor.

The secondary long assay retains nine peaks in 188/192 atlas conditions, versus 308/384 signed local conditions, 224/384 signed random conditions, and 100/192 shuffled-inward conditions. These correspond to 97.92%, 80.21%, 58.33%, and 52.08%; the signed methods have twice as many conditions because both signs are retained. This is *eventual* recurrence after settling, not a certificate of the immediate 300-unit signal. Atlas trials that eventually retain nine peaks still have immediate RMSE spanning 0.000491 to 0.875090. Preserving the regime and preserving its useful output are distinct requirements.

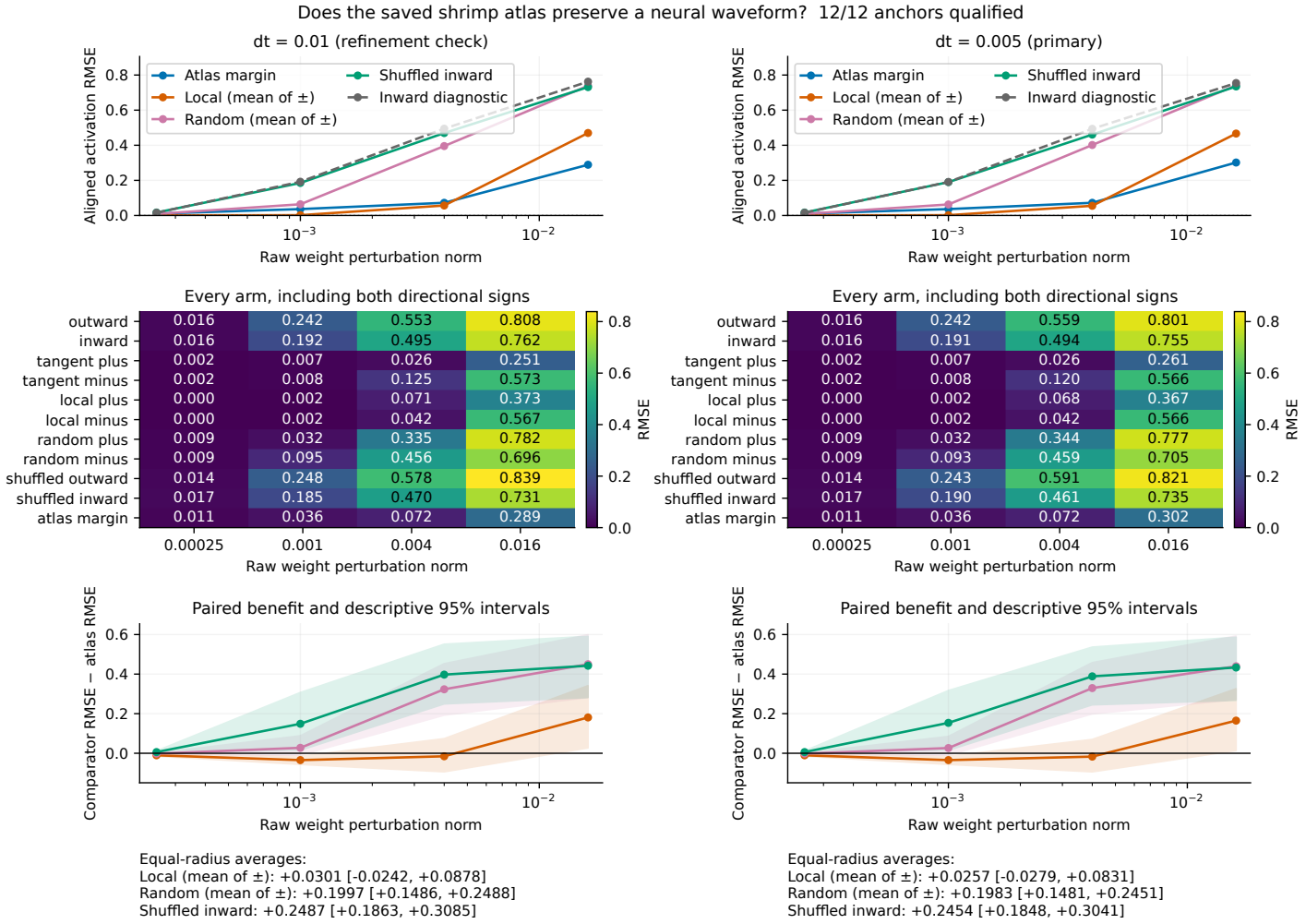


Figure 8: Measured utility depends on the size of the weight change. Columns retain both integration steps; 0.005 is the declared primary step. Top: mean waveform error for the atlas policy and paired competitors. Middle: all eleven direction arms at every declared dose. Bottom: comparator-minus-atlas error with descriptive anchor-bootstrap intervals. Small changes favor local sensitivity; the largest declared dose favors the atlas on average, while the pooled local comparison includes zero. All twelve anchors and 4,320 complete records contribute. Zero-dose numerical floors and numerical-step sensitivity remain visible.

6.6 The fixed example and numerical limits

The illustration was fixed before outcomes: the first qualifying upper-margin anchor, anchor 08, phase zero, radius 0.016. It favors the local method, despite the atlas's lower mean error at that dose. Its errors are 0.094559 for atlas, 0.011917 and 0.013725 for the two local signs, and 0.822093 for random plus. All four eventually retain nine peaks. Figure 9 retains this example rather than substituting a favorable atlas case. It shows why peak count alone cannot specify the signal delivered during a weight change. Transients, phase evolution, changed frequency and changed waveform can contribute; this assay does not isolate their individual causal contributions.

At step 0.005, zero-change shams have mean RMSE 6.5190×10^{-5} and maximum 6.9757×10^{-5}; none of the cadence diagnostics is flagged. All 2,160 coarse/fine pairs agree on eventual nine-peak retention, and 2,158 agree on exact detected peak count. Individual waveform errors are not uniformly step-converged: their absolute step difference has median 3.57×10^{-7}, mean 0.01550, 95th percentile 0.09142, and maximum 0.34511. The aggregate dose pattern survives: both steps favor local sensitivity at the smallest doses and the atlas at the largest dose, while both pooled local-comparison intervals include zero. This supports the reported finite numerical pattern, not precise step-independent outcomes for every trajectory.

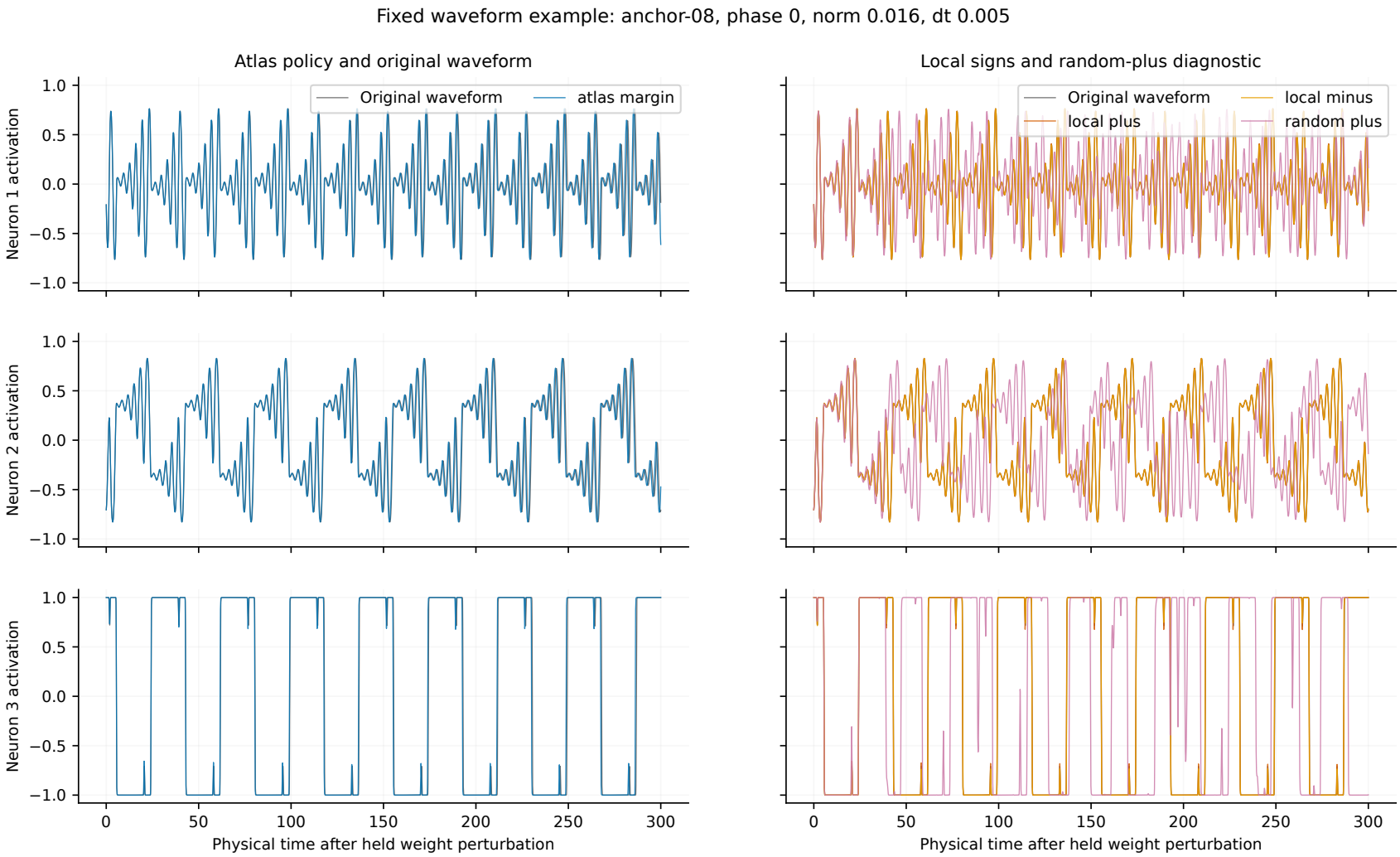


Figure 9: The actual three-neuron output after the same-sized weight change. The frozen illustration uses anchor 08, phase zero, radius 0.016, and step 0.005. Traces come from the immediate 300-unit probes; the original certified waveform is the reference. The local signs outperform the atlas in this particular example, preserving heterogeneity rather than illustrating only the average winner. The random-plus signal has large error despite eventual nine-peak recurrence in its separate long assay. No frequency refit or per-cycle phase alignment is used in the reported task score.

7 New starting weights test the proposed hybrid

EXP-007 freezes a follow-up after the full EXP-006 development results. Twelve new starting weights are selected from the same earlier atlas, excluding a closed radius-0.04 neighborhood of every development anchor in rectangle-normalized coordinates. The old atlas and its non-nine pixels remain unchanged. The remaining 3,758 eligible pixels are split into margin-rank thirds, with four starts per third; the smallest selected normalized distance from a development origin is 0.04006. This is a test of new waveform interventions conditional on an observed atlas, not generalization to an unseen map or network. Cross-stratum neighbors and overlap with old intervention destinations remain possible.

All twelve new starts qualify using independent common-initialization integrations at both numerical steps and unchanged primitive-orbit shooting gates. Four new phases use exact indices $337, 1012, 1687, 2362$ of each 2,700-sample endpoint-excluded certified orbit. The corresponding epochs are $iT/2700$; they are approximately, rather than exactly, the odd eighths of a cycle.

7.1 A stronger comparison with explicit preview access

The seven policies are atlas margin, signed local sensitivity, signed tangent, signed coordinate axes, uniform 16-direction search, augmented 21-entry search, and a hybrid. The hybrid uses the local method for radii through 0.004 and atlas margin at 0.016. This switch is chosen from EXP-006 before any new outcomes. Local sensitivity is integrated from the fixed initial state over 0–300, with its phase-projected Gram restricted to the scored interval below.

Each nonzero condition saves 21 separate simulator previews: 16 uniform directions, both local signs, both tangent signs, and the atlas direction. Duplicates are retained. Each preview starts from the same original orbit state, runs for 75 units at $\Delta t = 0.005$, and measures phase-aligned full-activation error. Local and tangent inspect two previews each, axes four, uniform search sixteen, and augmented search all twenty-one. The fixed atlas requires no preview. Augmented search therefore uses geometry and local derivatives; uniform search is the atlas-free search control. All policies use the same native weight-change norm. Their information and compute costs differ.

Policy choices are fixed before independent 300-unit target trajectories at both integration steps. The primary metric scores only the 2,250 samples $75.1 \le t \le 300$, fitting one constant phase shift within that interval. The preview and scoring samples are disjoint but dynamically dependent. Clonable simulator access is an explicit assumption. Full 0–300 error is secondary. Neither frequency refitting nor time warping is allowed. Equal phase, dose and anchor weighting, missingness rules, and 10,000 descriptive paired-anchor bootstrap draws are frozen in advance. The hybrid and local policy coincide at the three smaller doses, so their pooled contrast is exactly one quarter of the largest-dose contrast; these are not separate demonstrations.

7.2 The new comparison does not establish a general hybrid advantage

All 4,032 preview entries and 2,784 target records complete, with no replacement or missing outcome. Table 6 reports the primary tail error. At radius 0.016, hybrid mean error is 0.115448, versus 0.260336 for signed local sensitivity. The mean paired reduction is 0.144887, but its descriptive 95% interval spans $[-0.045524, 0.353194]$ and only six of twelve anchors favor hybrid. The point estimate is 55.65% lower, not a resolved advantage across these starts. Augmented search has a smaller largest-dose mean, 0.076644; its paired comparison with hybrid is also unresolved.

At the smallest two radii, local/hybrid beats the tangent and coordinate rules at every anchor. At radius 0.004, however, its mean error is 0.132630, versus 0.027130 for tangent and coordinate

policies. Thus the frozen switch fails to capture an intermediate-dose opportunity. This observation does not authorize moving the switch and claiming success on the same data. The pooled hybrid mean is 0.062373, compared with 0.098595 for local, 0.051095 for tangent/axes, 0.074171 for uniform search and 0.049246 for augmented search. Every one of the five primary pooled intervals includes zero (Table 7).

Radius	Atlas	Local	Tangent	Axes	Uniform	Augmented	Hybrid
0.00025	0.013149	0.000244	0.002412	0.002412	0.001440	0.000580	0.000244
0.001	0.056555	0.001169	0.008732	0.008732	0.005639	0.002351	0.001169
0.004	0.135219	0.132630	0.027130	0.027130	0.107517	0.117410	0.132630
0.016	0.115448	0.260336	0.166105	0.166105	0.182088	0.076644	0.115448
All doses	0.080093	0.098595	0.051095	0.051095	0.074171	0.049246	0.062373

Table 6: EXP-007: all seven frozen policies. Mean all-neuron activation RMSE on $75 < t \leq 300$, primary $\Delta t = 0.005$. Uniform uses sixteen previews; augmented uses twenty-one entries including local and atlas information. All doses are equally weighted. The table retains unfavorable doses and identical-action controls.

Comparator	Largest-dose benefit [95% interval]	Pooled benefit [95% interval]
Local	0.144887 $[-0.045524, 0.353194]$	0.036222 $[-0.011381, 0.088298]$
Tangent	0.050656 $[-0.043143, 0.153512]$	-0.011278 $[-0.050720, 0.028091]$
Axes	0.050656 $[-0.043143, 0.153512]$	-0.011278 $[-0.050720, 0.028091]$
Uniform 16	0.066640 $[-0.042759, 0.201205]$	0.011798 $[-0.028640, 0.052582]$
Augmented 21	-0.038805 $[-0.097362, 0.031163]$	-0.013127 $[-0.033383, 0.008182]$

Table 7: All declared comparator-minus-hybrid contrasts. Positive values favor hybrid. Intervals resample twelve selected anchor clusters and are descriptive; phases and integration steps are not independent replications. All ten displayed intervals include zero. No resolved overall advantage or equal-total-cost benefit is established.

7.3 A coordinate control explains the tangent result in this sample

The tangent and signed-axis policies choose exactly the same actions in all 192 nonzero conditions: $+\gamma$ in 168 and $-\gamma$ in 24, always holding α fixed. All 192 corresponding preview state arrays and all 384 target state arrays across both steps are identical. The old nearest-pixel normals happen to be horizontal at all twelve selected starts, so their tangents are coordinate directions. This equivalence is a result of the retained action records, not a renaming of a comparator after outcomes. Tangent performance here therefore supplies no independent evidence that detailed shrimp geometry is necessary; a simple signed coordinate test realizes the same actions. It does not establish that changing γ is always best or that geometry has no value elsewhere.

The preselected example also prevents an exclusively favorable illustration. At the first eligible start and phase, largest-dose hybrid error is 0.059668, local error 0.037763, and both uniform and augmented search error 0.589753. The augmented selector minimizes its preview objective but does not minimize later waveform error. The example is retained in Figure 12; its ranking differs from the aggregate largest-dose ranking. More previews are not automatically a better predictor when the selection and evaluation horizons differ.

7.4 Numerical and cost limits

All 48 fine-step zero-dose shams complete, with mean tail error 6.53×10^{-5} and maximum 6.88×10^{-5}. Across 1,392 paired coarse/fine targets, the absolute RMS difference has median 1.30×10^{-7}, mean 0.00403, 95th percentile 0.01456 and maximum 0.23486. The coarse results preserve the unresolved pooled-comparison conclusion, but these differences rule out a claim of uniform pointwise convergence.

Collection and summary construction take 137.51 seconds with four local workers, excluding final verification. Forty-eight sensitivity solves consume 23.52 seconds within that total. Summing the overlapping preview-call wall times gives 161.85 seconds; that is neither elapsed duration nor CPU time. The old atlas costs a separate 140.54 seconds, and per-policy preview counts range from zero to twenty-one. Shared research execution does not make these information sources free. All 7,149 receipt-listed artifacts are retained. Independent interpolation reconstructs 504,320 saved RMS values across 9,600 metric blocks with maximum absolute discrepancy 4.44×10^{-16}. No paid resource is used.

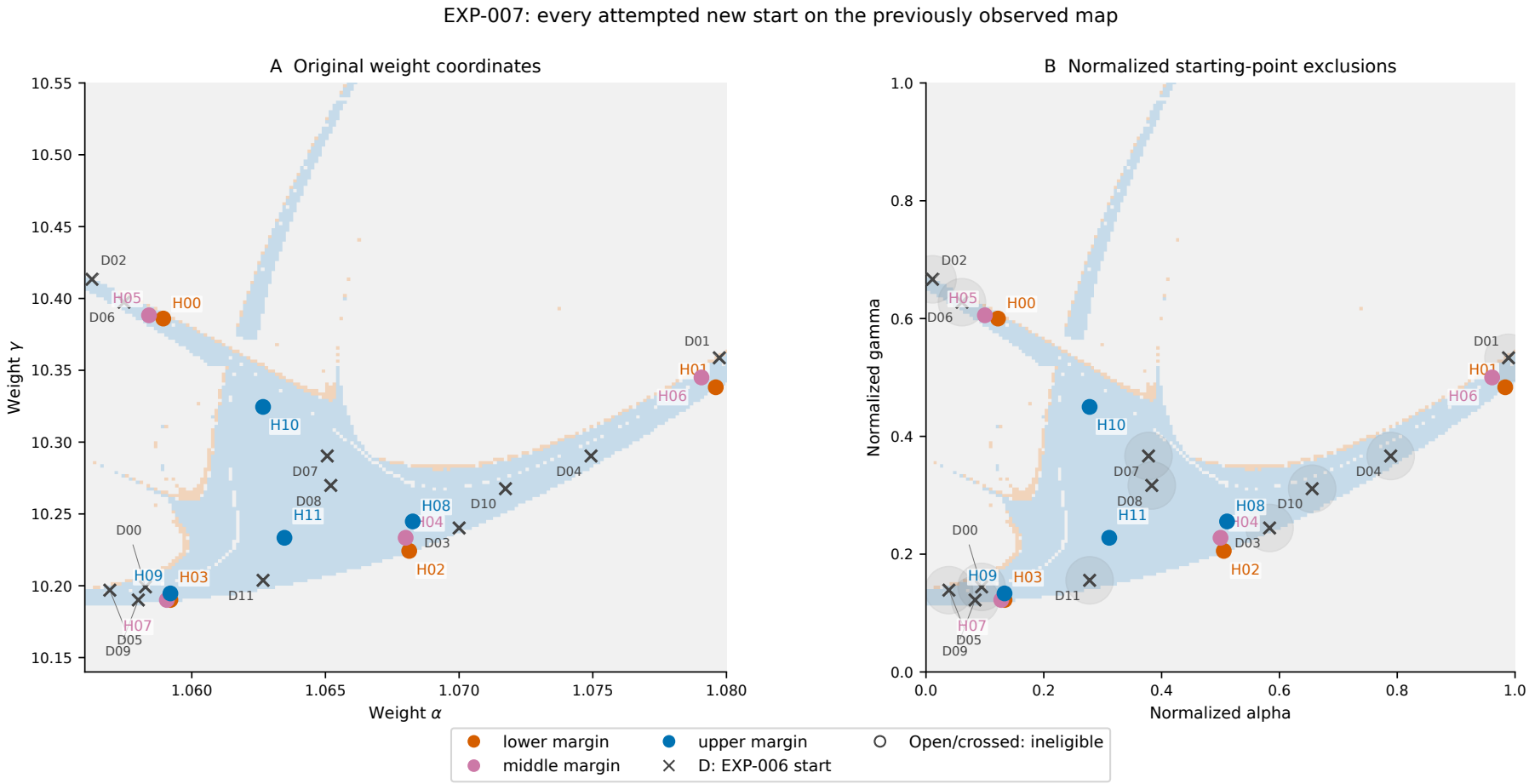
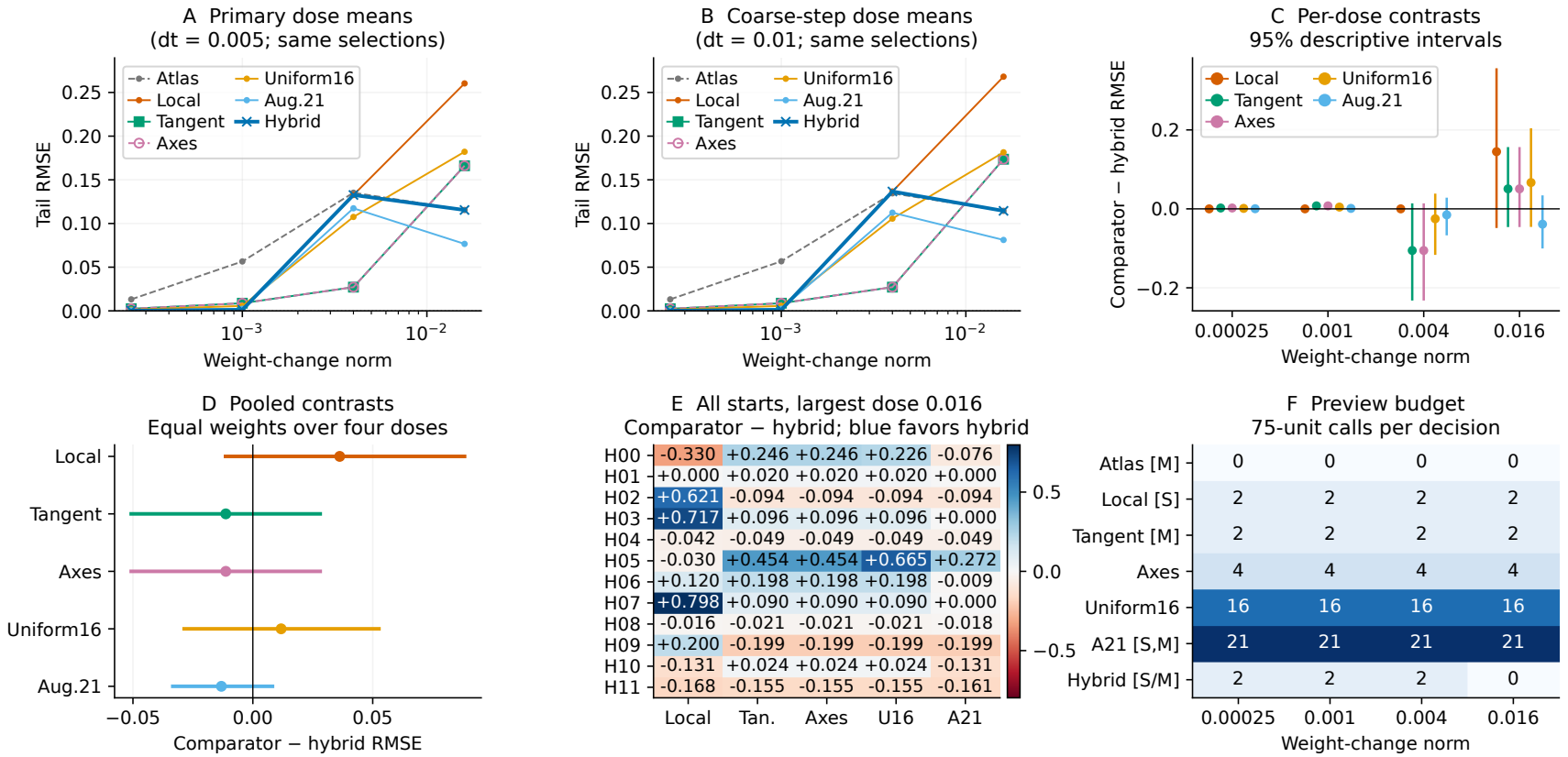


Figure 10: Where the new interventions start. The measured EXP-004 period atlas underlies the new and development anchor locations. The second view expresses coordinates in units of the focus rectangle widths so the fixed radius-0.04 exclusion neighborhoods can be shown as circles. The exclusion applies to starting locations, not every previous destination or trajectory. All twelve attempted new starts are retained. The same previously observed atlas supplies margins and tangents; the design does not test a new network or an unseen parameter map.



Tail: 75<t≤300. Positive contrasts favor hybrid. Tangent/axes select identical actions here.
Intervals are descriptive for selected anchors; no population/multiplicity claim.
H00–H11: holdout-00–11; Tan.=tangent; U16=uniform16; A21=augmented21. Gray/×: undefined; no success-only deletion.
[S] sensitivity; [M] old map. Hybrid uses S for small doses, M for largest. All require baseline; previews shared in collection.
Collection 137.51 s; preview-call sum 161.85 s; 48 sensitivity calls 23.52 s. Concurrent sums ≠ elapsed time.
Old atlas: 140.54 s separately. Preview counts exclude prerequisites. Total compute is not matched; no speedup established.

Figure 11: New-start policy comparison with stronger controls. Dose means at both integration steps score only $75 < t \leq 300$. Fine-step preview choices are reused at both steps. Positive comparator-minus-hybrid contrasts favor the hybrid; all five declared comparator contrasts are retained, including the largest-dose per-anchor results. Intervals are descriptive paired-anchor bootstrap percentiles, not population confidence claims. The preview budget panel counts candidate calls, not equal total resources: atlas construction and local sensitivity are additional dependencies. Augmented 21-entry search includes atlas and local directions; uniform 16-direction search does not. The hybrid copies local at three doses, making those local contrasts identically zero.

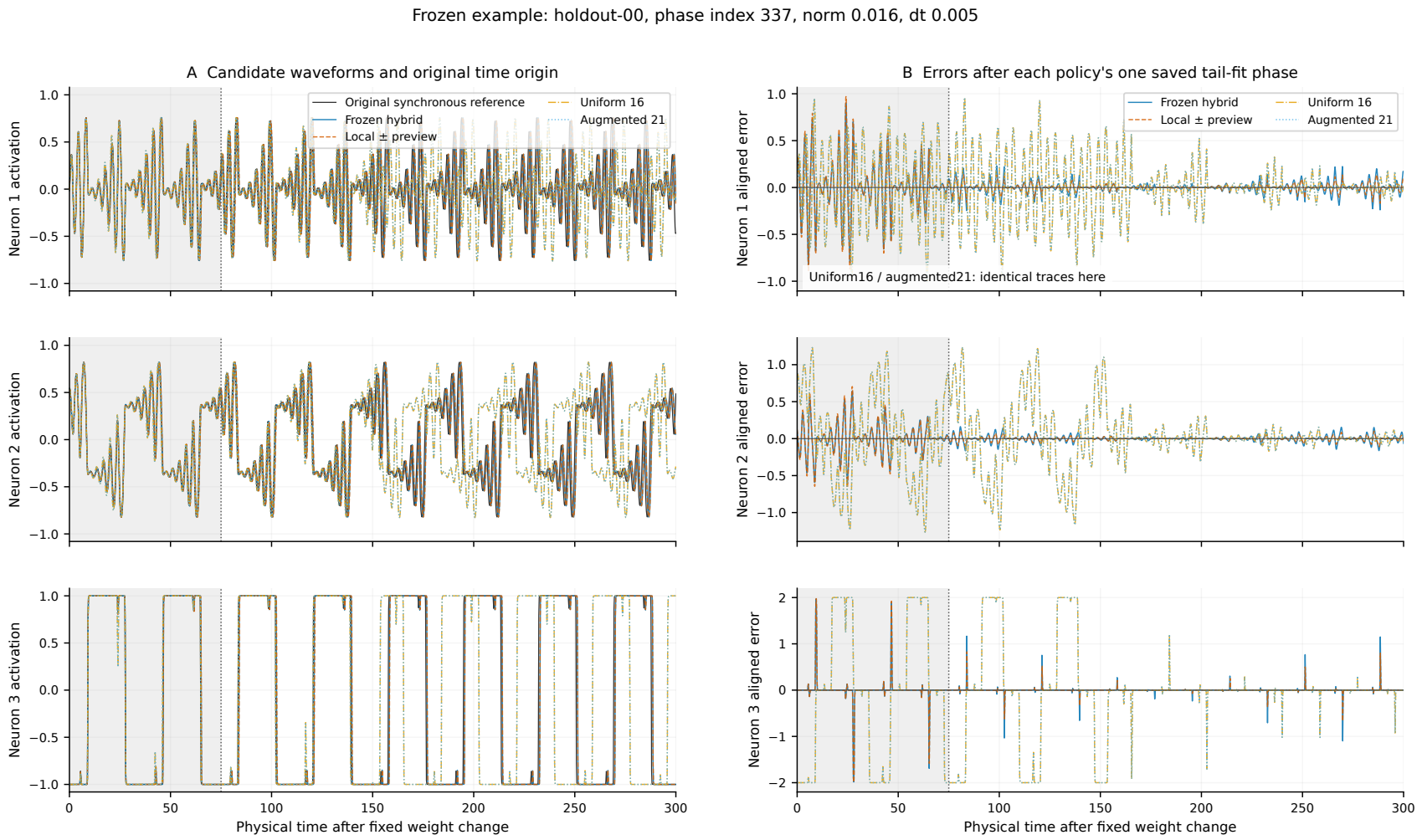


Figure 12: A prospectively selected new-start waveform example. First eligible anchor, first phase, largest dose, and primary numerical step. Rows show actual activations and componentwise errors. Preview time is shaded; the primary endpoint excludes that interval. Candidate curves are shown against the original synchronous reference, while each error curve uses its own single phase shift fitted to the entire scored tail. No frequency correction or separate cycle alignment is used. This example follows the frozen selection rule even if it is unfavorable to the hybrid or unrepresentative of the aggregate.

8 What this means for managing the network

The shrimp maps particular neural rhythms and their local attraction. Continuation shows that an orbit can survive while losing attraction; matched interventions show that eventual periodicity can coexist with immediate waveform error. EXP-006 supplies a development-stage largest-dose advantage over its declared local comparator. EXP-007 tests stronger methods at new starts and leaves all primary pooled comparisons unresolved. Its tangent rule is exactly reproduced by a coordinate control. Thus the map is a description of meaningful network behavior, while its incremental value for managing that behavior remains a separate, unresolved question. Maximizing pixel margin is not an established signal-preservation objective.

Whether to seek or avoid a periodic region therefore depends on the intended computation. A rhythm generator can use a stable periodic operating region, subject to waveform and timing requirements. A solver intended to settle to one answer can fail by cycling. The present waveform task provides a measured small-network example; it does not resolve the corresponding decision for training or reasoning systems.

The next experiments should isolate the remaining functional questions:

1. **Test whether geometry improves a task forecast.** Keep EXP-006 and EXP-007 as development evidence. On fresh intervention blocks, compare prediction of later waveform/phase error using local derivatives, short previews, and those same inputs plus recurrence/Floquet information. Use identical candidate actions and total resources. The immediate question is whether geometry adds information beyond the inexpensive coordinate and local controls, not whether another switch can be fitted to these anchors. Retain preview-to-later ranking failures.

2. **Separate stable rhythm retention from useful signal retention.** For fixed prospective points spanning the verified window and its boundary, measure late waveform, period/frequency, transient recovery and state-disturbance response separately. Refine the recorded continuation crossings and failed checkpoint in a new attempt while preserving the original failures. Pointwise errors with substantial step disagreement require independent numerical checks before controller conclusions.

3. **Test a controller only after an incremental benefit survives.** Freeze a sequential disturbance-and-action task, waveform/phase tolerance, recovery metric and effort budget. Compare geometry as a feasibility constraint with direct task-quality control. A multiple-memory extension could separately test spectral diversity and crosstalk. Learned-model transfer follows only after a reproducible task benefit, not from the shrimp's visual prominence.

The present map reproduces recognizable shrimp-shaped organization and verifies selected neural orbits; it does not certify every pixel or the exact boundary topology. Compared with the published 1000×1000 scans, $\Delta t = 0.01$, and settling time 10,000, our scan has lower resolution and shorter settling. Source initial states and row-reset conventions were unspecified; ours are explicit. The printed source transect equation appeared inconsistent with its displayed range, so we used the independently frozen vertical transect. These differences delimit reproduction rather than being hidden numerical substitutions.

8.1 A separate clue: interference between dynamical memories

The unreviewed preprint of Aguilera and De Martino [1] proposes that spectral diversity reduces interference between encoded dynamical memories in a large stochastic network. Its retrieval

criterion measures overlap magnitude, not waveform fidelity; its extensive-capacity limit does not mean unlimited finite-network storage. This motivates a separate multiple-memory hypothesis, not evidence for shrimp geometry or our controller.

Our analytical audit also finds a printed inconsistency: for independent zero-field spins refreshed with probability Δ, the update law gives $q_{s+\ell,s} = (1-\Delta)^{\ell}$ and $\chi_{s+\ell,s} = \beta\Delta(1-\Delta)^{\ell-1}$, whereas printed supplemental equations S40–S41 omit persistence or refresh factors. Other formulas retain those factors, so implementation consequences remain unknown. Moreover, a finite strictly causal response matrix is nilpotent; the paper's response-radius criterion needs an explicit infinite-time operator definition. The versioned reading note and exact independent-spin tests are retained in `docs/literature/`. These checks do not reproduce its capacity results.

Evidence, cost, and research status

The code, protocol, and machine plan were pushed at revision `b670086775bb` before EXP-004 outcomes. Its one completed attempt used four local CPU workers and 140.54 seconds of measured wall time. EXP-005 was frozen and pushed at revision `01209371d541` before its eight fresh runs; its completed attempt used one worker and 4.17 seconds. Recorded paid compute and paid review costs are zero. Raw row states, peak events, statistics, selected probes, refinements, and certificates are retained with source hashes. Figures retain plotted arrays and source/output receipts. The plotting tools render saved data without advancing the ODE. All protocols are in `docs/`, and exact plans are in `experiments/hopfield/`. Complete attempts are retained under `evidence/EXP-004/`, `evidence/EXP-005/`, `evidence/EXP-006/`, and `evidence/EXP-007/`. Private Git history supports internal chronology, not public preregistration.

EXP-006 was frozen and pushed at revision `ddb568e6cd67` before new trajectories. Its one complete attempt used four local trial workers and 423.87 seconds through collection and summary construction, excluding final verification. The 48 local-sensitivity solves consumed 26.63 seconds within that total. The earlier atlas's 140.54 seconds are a separate information cost. No net speedup or equal-budget controller advantage is claimed. All 4,472 receipt-listed artifacts are retained, and a separate audit reproduces all 4,320 recorded waveform scores with zero discrepancy at the saved phases. Numerical source and selection remain frozen; a documented post-run display/verification amendment changes only figure readability and floating-point replay portability. The EXP-004–006 figure families remain unchanged, with their exact plotted arrays and receipts. Paid compute and paid review remain zero.

EXP-007 was frozen and pushed at revision `ac098c3e3833` before its own outcomes. Its complete source and the earlier experiments' source commits remain under archive tags. The expanded manuscript contains twelve empirical plates with 59 measured panels and seven numerical tables. The separate preprint audit is analytical; its capacity numerics have not been reproduced here.

Generative-AI tools assisted with design, implementation, numerical execution, audits, plotting, and drafting. The human selected the question; authorship and final release approval remain pending. No paid Pro review was requested or run.

A Earlier pilot history

EXP-001–003 remain unchanged and are not evidence for the neural shrimp reproduced here. The Hénon control verified nontrivial attracting cycles in 39 of 65 cells. The tiny tanh *training* scan found 0 verified cycles in 162 cells, with 107 unresolved and 55 divergent. In the 96-run policy pilot, the recurrence detector never intervened; each policy had 9 divergent cases among 24 pairs. The

old plots, exact scalar toy, and receipts remain in the repository as exploratory history. EXP-004 supplies a separate, measured recurrent-network periodic window; it does not retroactively validate those training hypotheses.

References

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